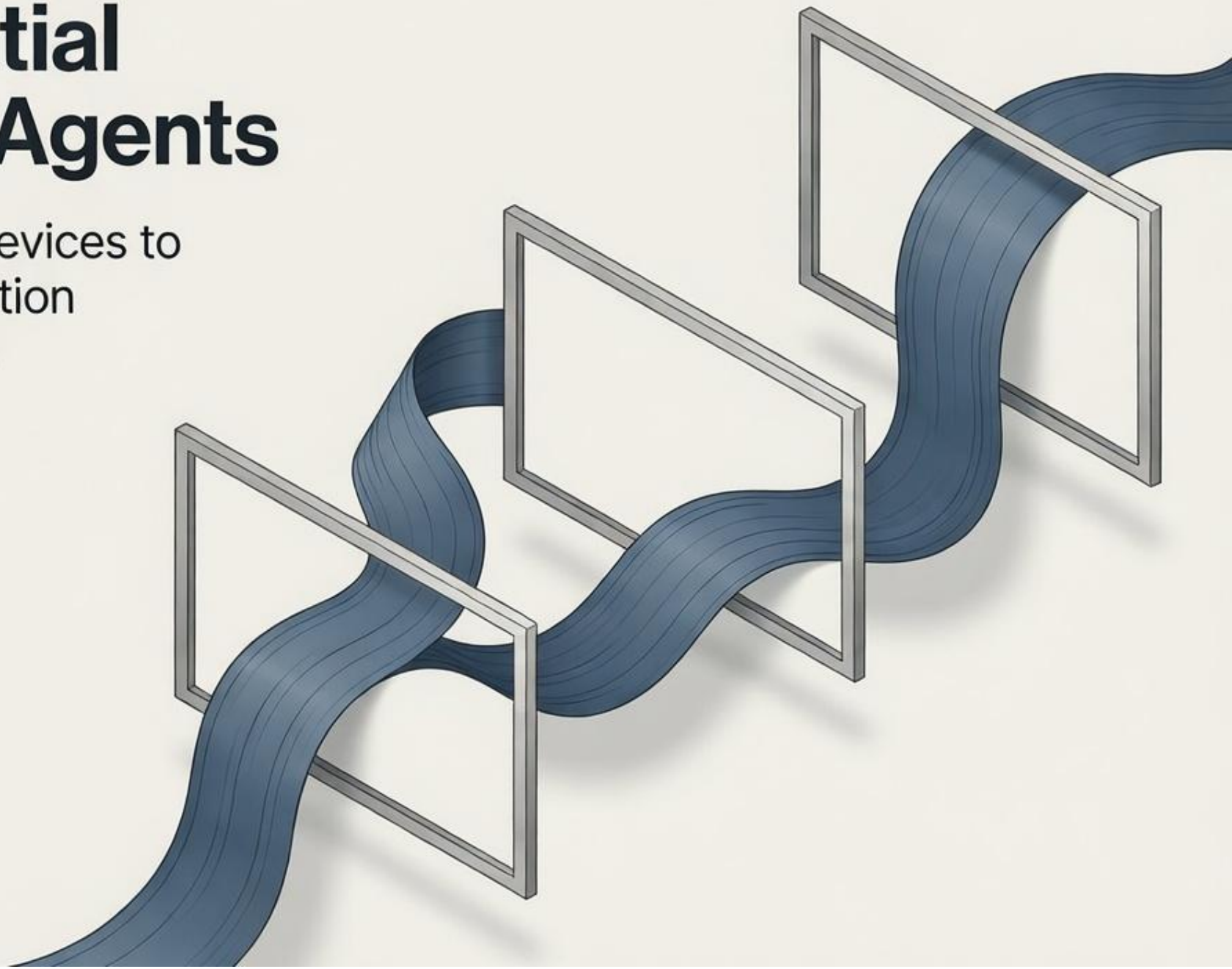
AI2X: The Spatial Canvas for AI Agents

Decoupling displays from devices to create a dynamic orchestration layer for the physical world.

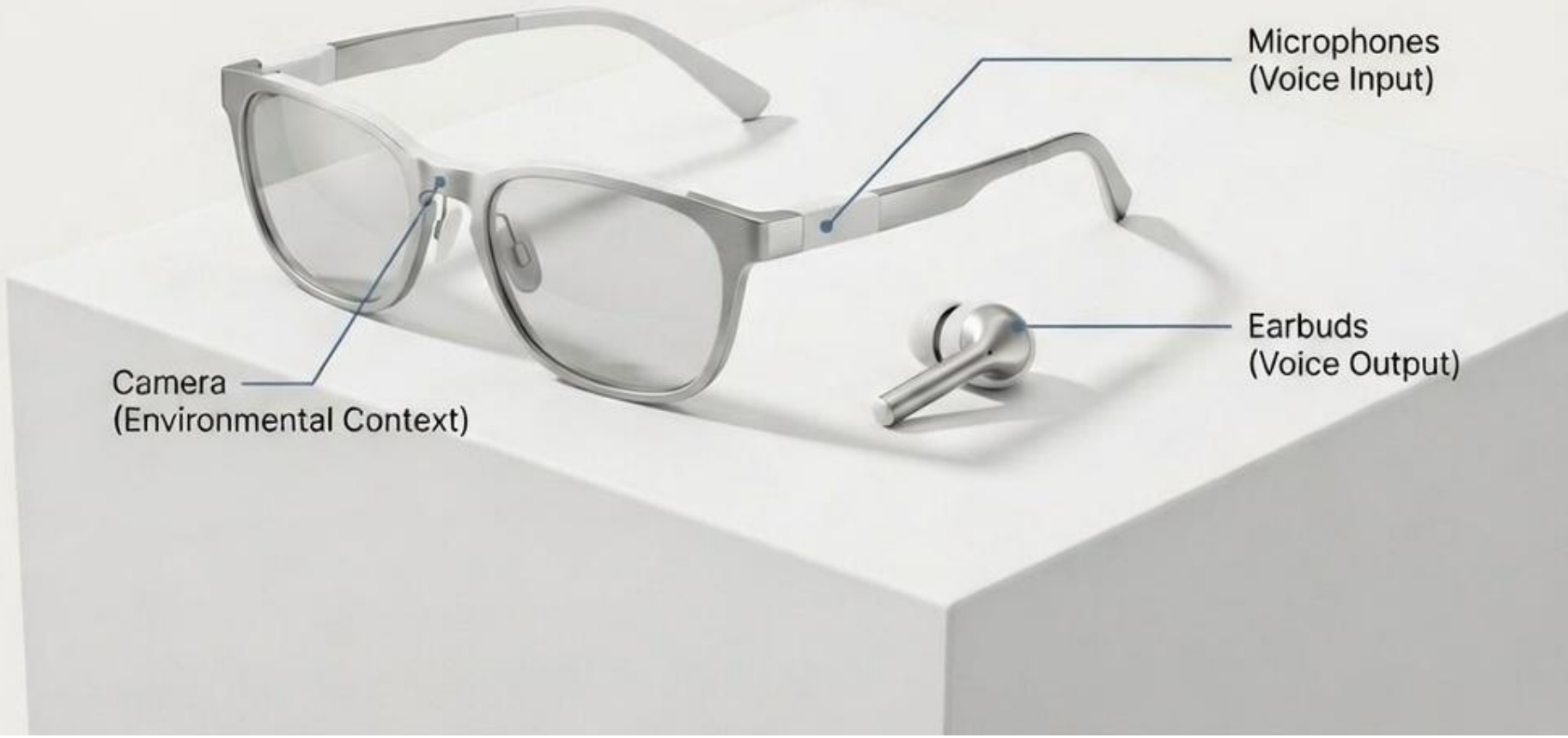


The future of AI hardware is lightweight and voice-first

Future wearable devices will abandon heavy, power-hungry display modules.

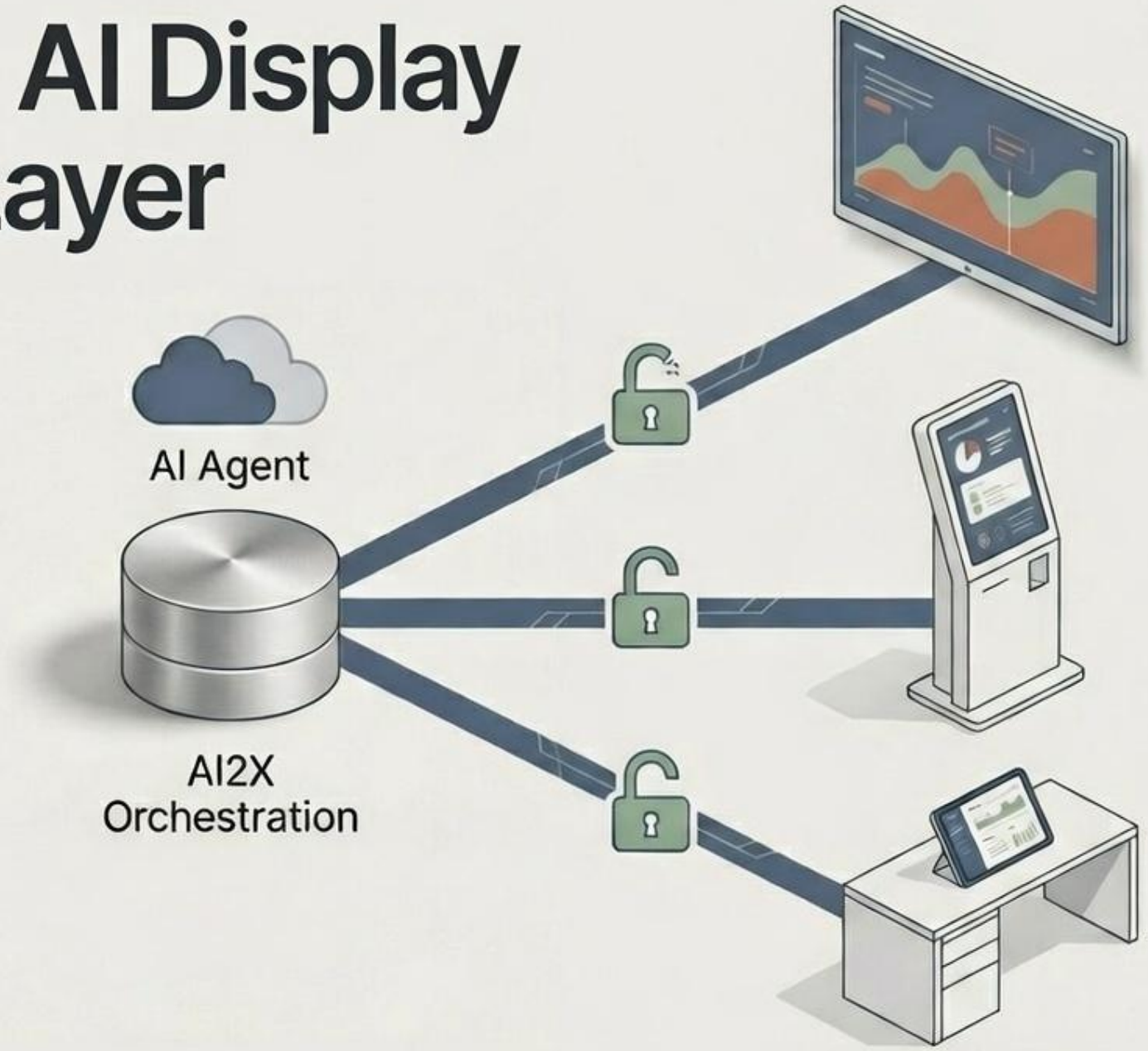
Most daily AI interaction will seamlessly mirror human conversation.

The hardware focuses solely on capturing visual context and maintaining a continuous audio link.

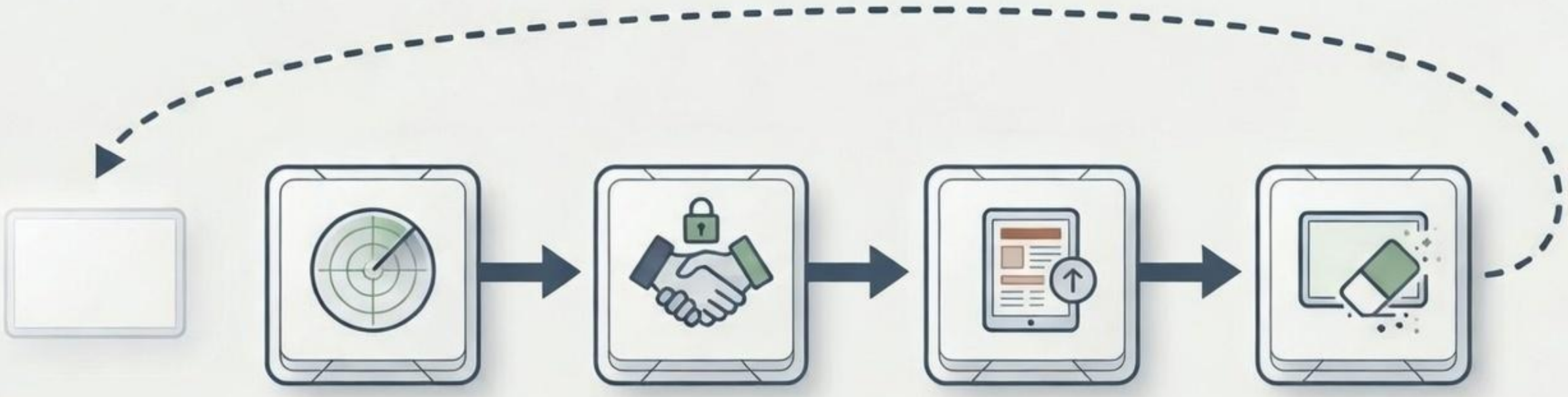


Introducing the AI Display Orchestration Layer

- AI2X is an infrastructure layer, not a physical device.
- It allows AI Agents to dynamically discover available screens, temporarily claim them, render structured information, and instantly release them.
- Screens are no longer personal devices; they become a shared, temporary canvas for AI.



The AI2X Session Lifecycle



Neutral State

1. Discovery

Agent queries the Multi-Display Control Plane (MCP) for available, policy-compliant displays.

2. Claim

Session established. AssignmentId generated. TTL (Time-to-Live) lease activated.

3. Render

Deterministic templates push structured data to the screen. Optional 'Renew' if the task extends past TTL.

4. Revoke

Task complete. Session terminated. Memory cleared. Display instantly restores to its Neutral State.

A three-tier architecture decoupling reasoning from display

Top: User & AI Agent Layer

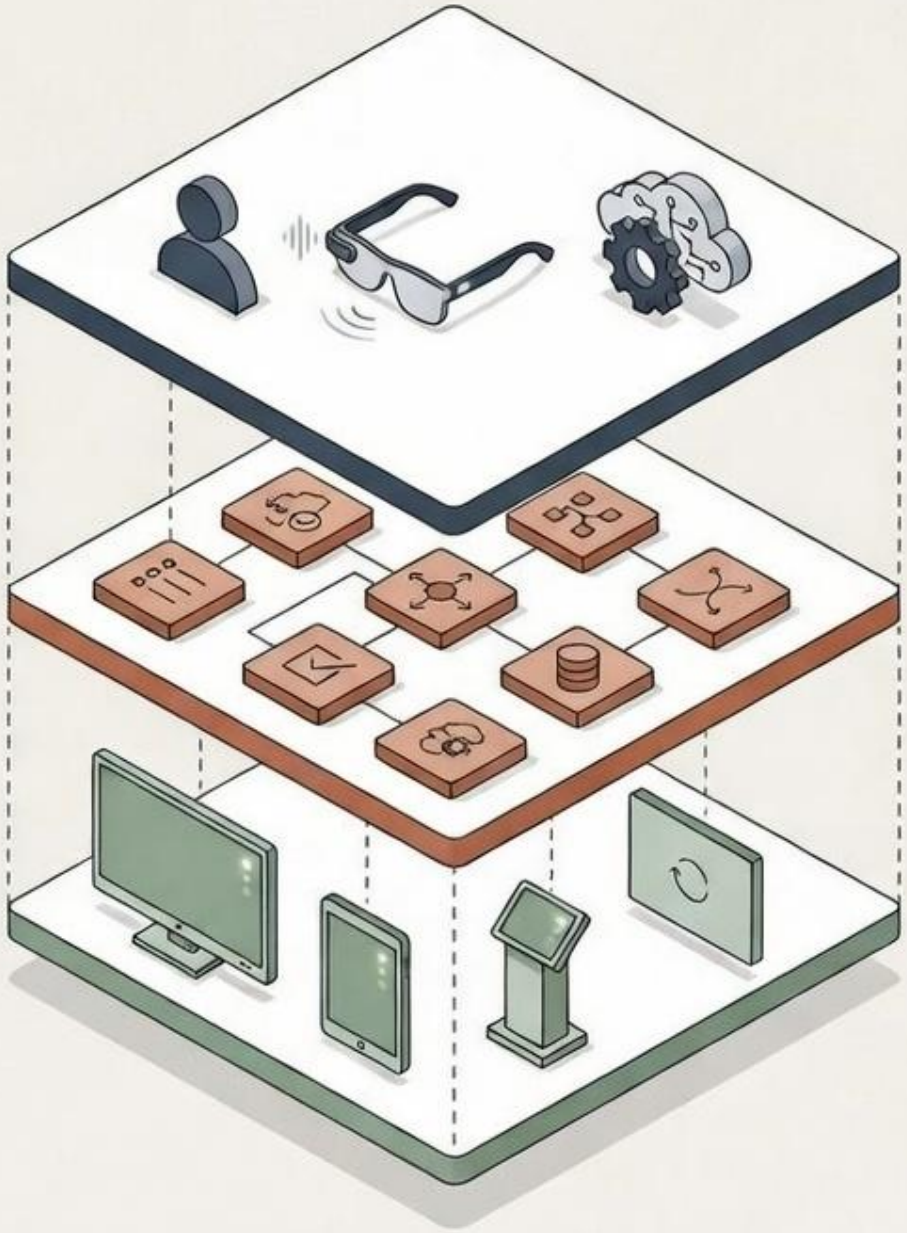
Wearable Interface (Voice/Context), Agent (Reasoning, Planning, Tool Calling).

Middle: AI2X Orchestration Layer

Policy Engine, Template Validation, Multi-Display Control Plane (MCP) managing Display Registry and Leases.

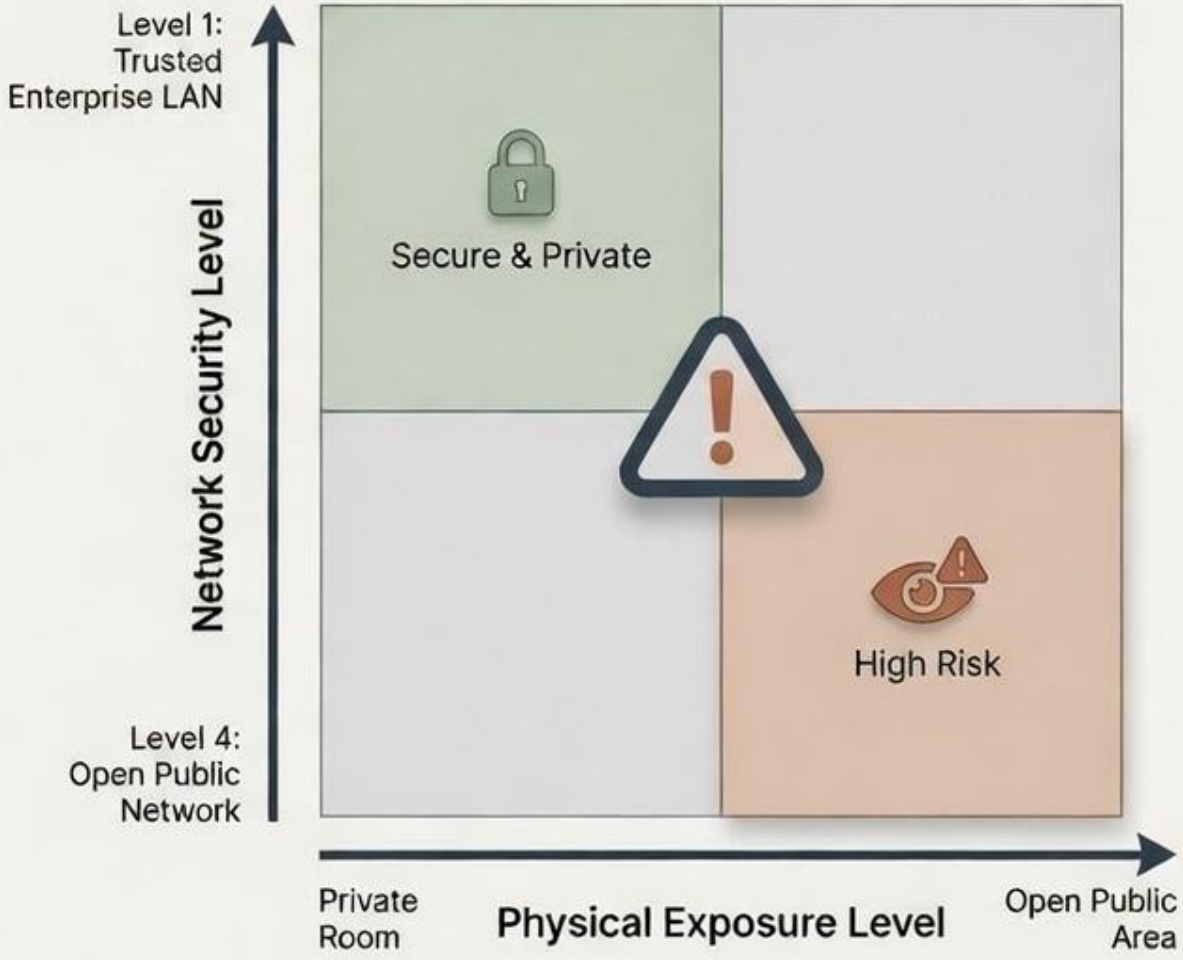
Bottom: Display Endpoints Layer

Display Gateways maintaining WebSocket connections, pushing to Runtime Displays (TVs, Tablets, Kiosks) and ensuring Neutral State restore.

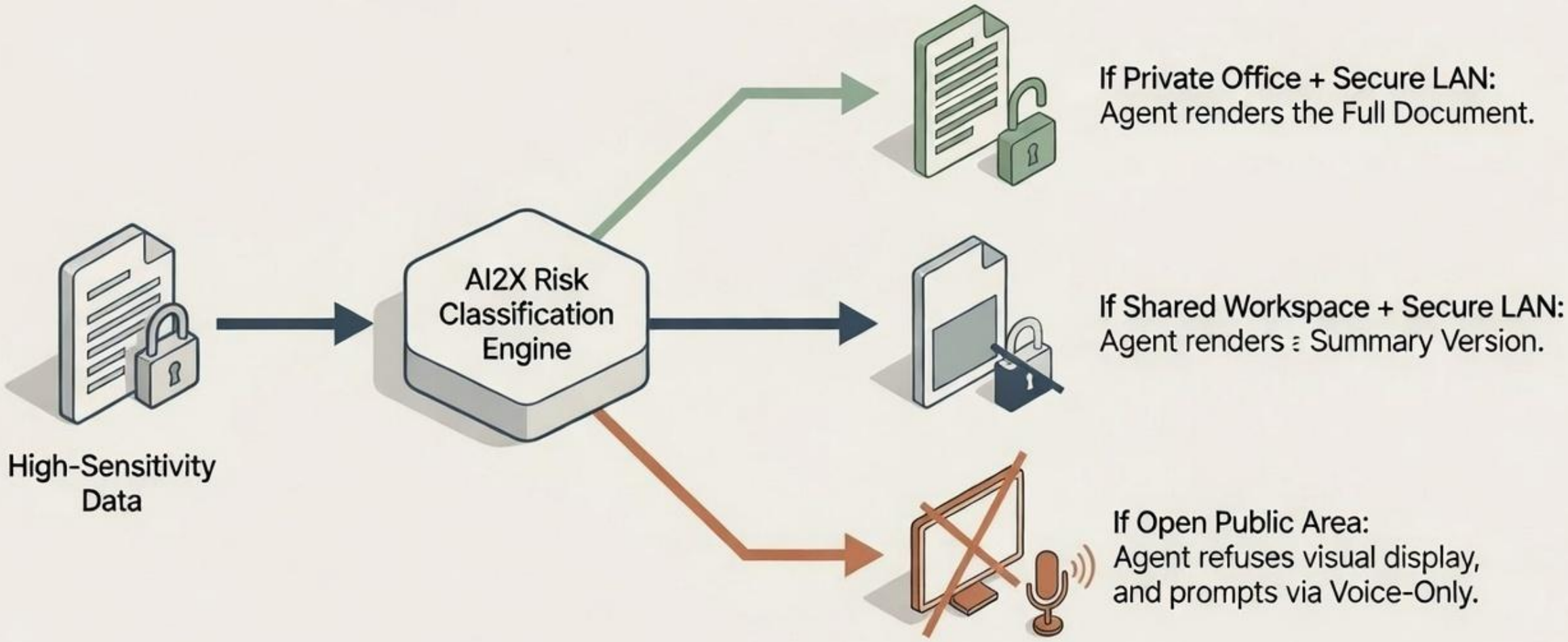


The Spatial Security Challenge: Classifying physical risk

- AI2X assumes all displays are semi-public environments by default.
- Security cannot rely solely on encryption; it must account for physical visibility.
- Every display is classified on these two vectors before any data is rendered.



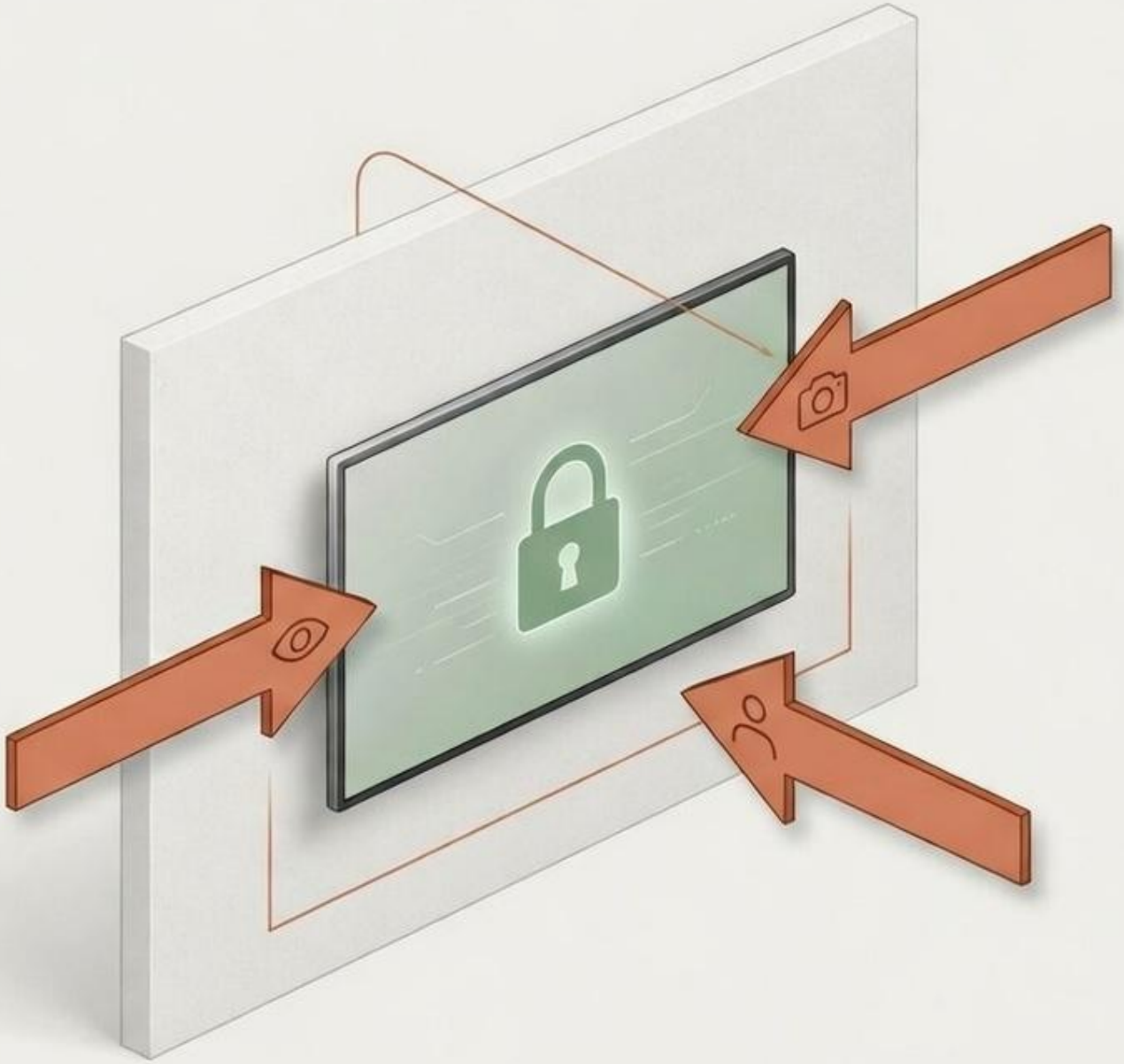
Equipping AI Agents with Spatial Risk Awareness



AI2X explicitly defaults to caution, giving software the context needed to navigate physical-world risks.

The structural gap in open environments

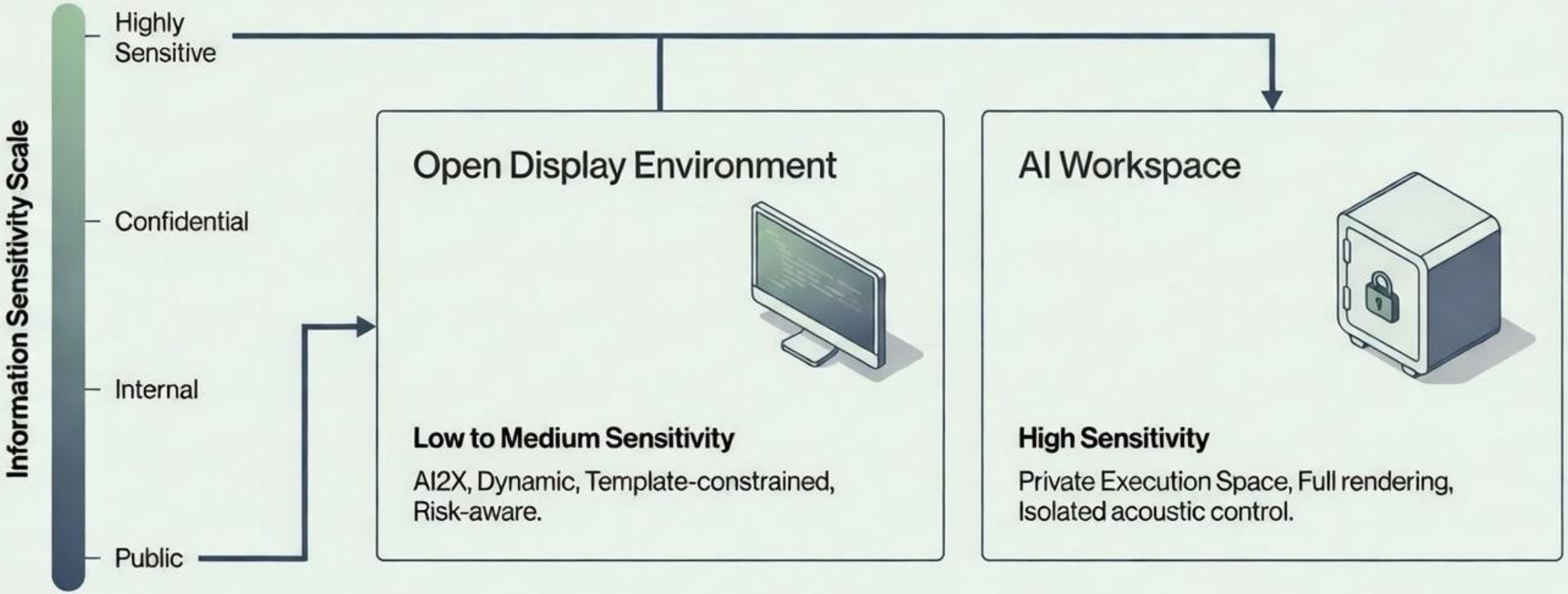
- Despite rigorous software governance and ephemeral memory, AI2X cannot eliminate the physical unpredictability of the real world.
- Shoulder surfing, physical camera recording, and accidental room entries remain structural risks for highly sensitive enterprise data.
- Software orchestration requires a physical counterpart for maximum-security tasks.



Introducing the AI Workspace

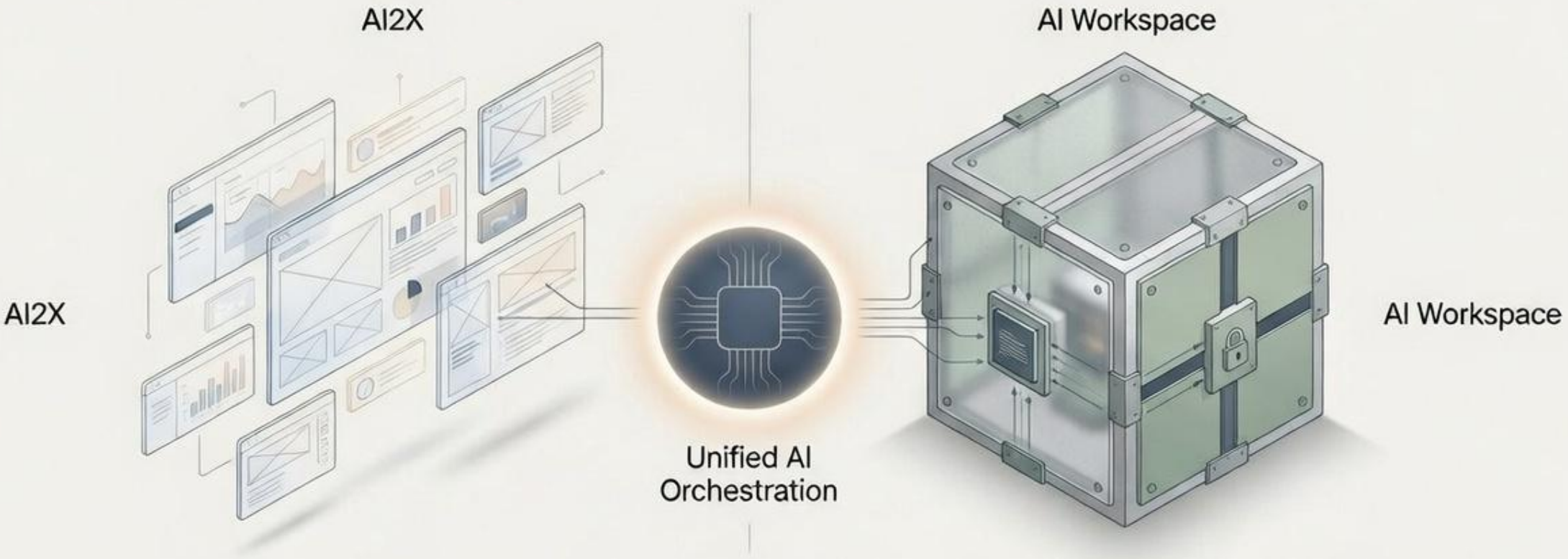
A closed, secure control room designed exclusively for high-sensitivity AI execution.

A layered spatial security architecture



AI Agent determines sensitivity → AI2X enforces policy → Content is routed to the appropriate physical execution layer.

The complete spatial computing infrastructure



AI2X provides the dynamic canvas.

AI Workspace provides the secure control room.

Information is structured. Sessions are controlled. Data is transient.

AI is here. Anywhere.